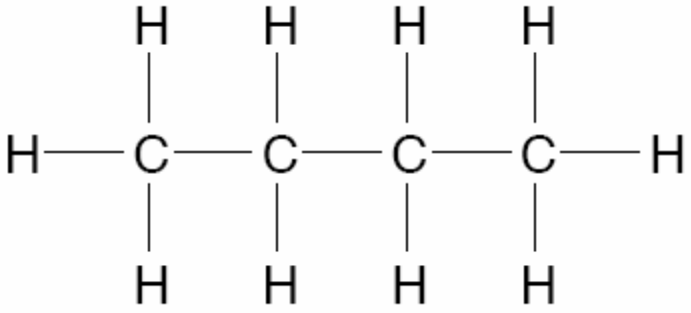
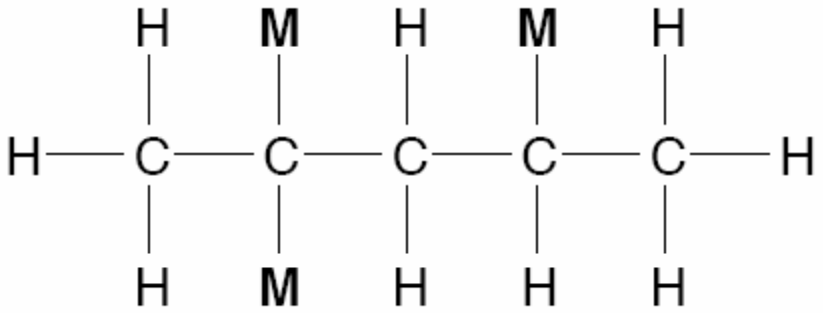
Combustion：

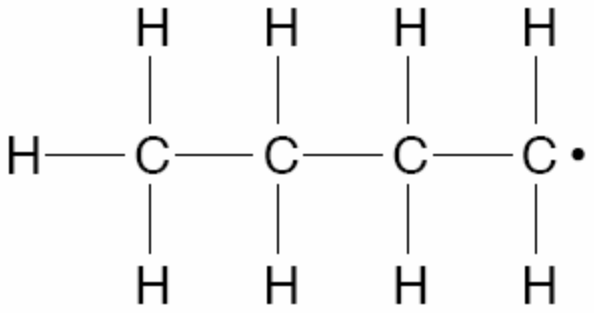
Chemical Kinetic Mechanisms for Practical Fuels

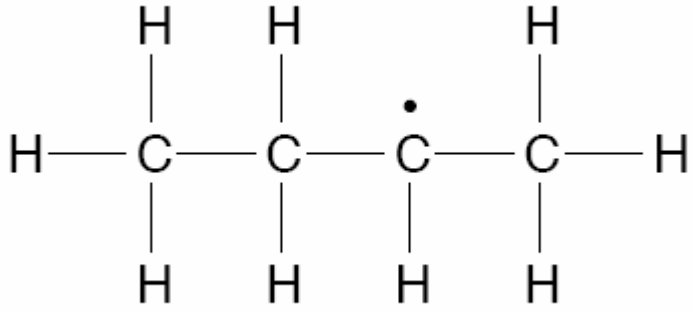
Instructor: Zhuyin Ren, Haitao Xu

Review of Fuel Names： Alkane, Alkyl

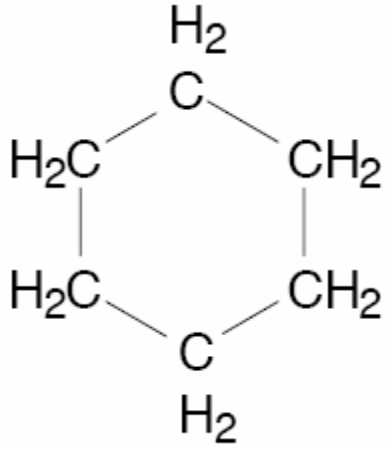


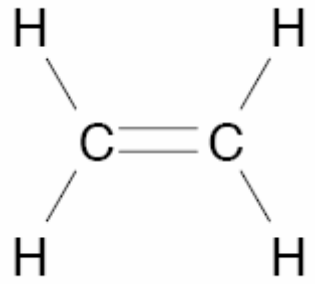


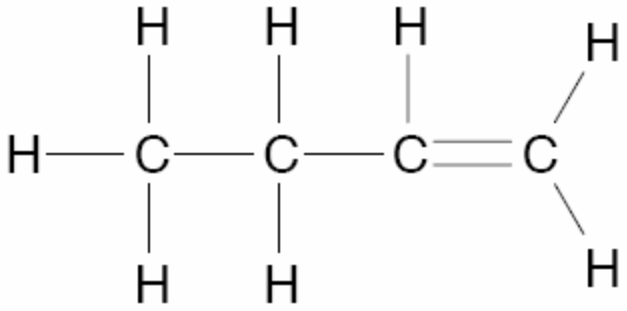


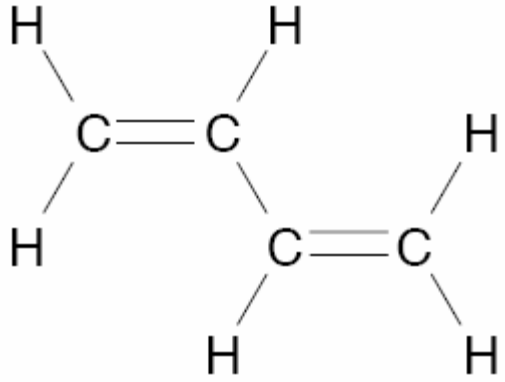


Cyclo-alkane, Alkene

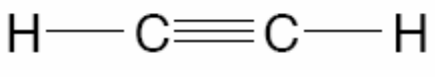


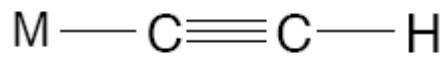


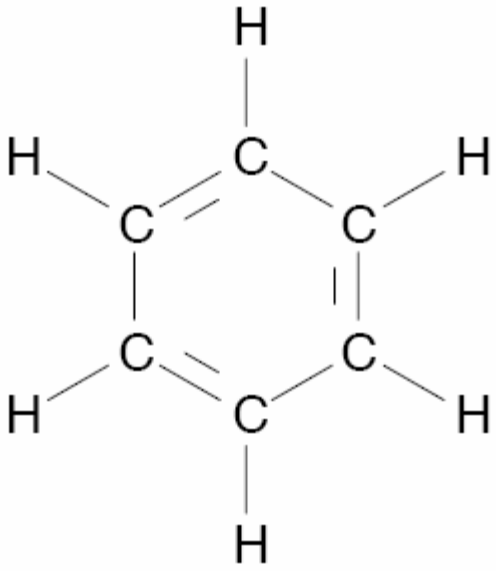


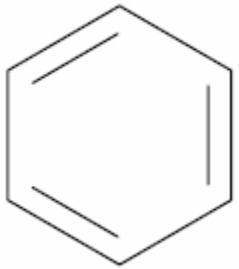


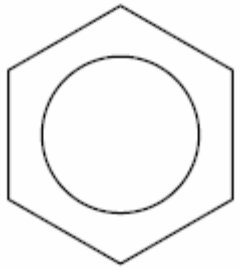
Aromatics, Alkyne

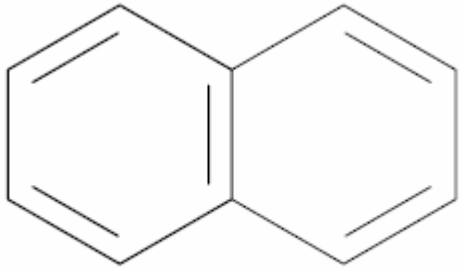






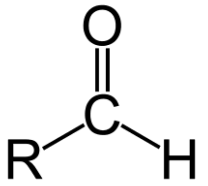




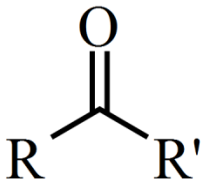


Other Types of Fuels

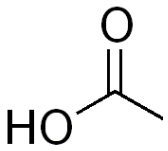
- Alcohols, (-OH)
  - Methanol (CH3OH), ethanol (C2H5OH), …
- Aldehydes (-CHO)
  - Formaldehyde (HCHO, or CH2O)
  - Acetaldehyde (CH3CHO)



- Ether (-O-)
  - Dimethyl ether (DME) CH3-O-CH3
- Ketone (-CO-)
  - Methyl Ethyl Ketone (2-Butanone)



- Fatty Acid (-COOH)



- …

H2 – O2 Explosion Limits

First limit: balance between radical creation and loss to the wall

Second limit: balance between chain branching and chain termination

R1: $$H + O_2 \rightarrow O + OH$$

R2: $$O + H_2 \rightarrow H + OH$$

R3: $$H_2 + OH \rightarrow H_2O + H$$

R9: $$H + O_2 + M \rightarrow HO_2 + M$$

Third limit: chain termination becomes branching

$$HO_2 + H_2 \rightarrow H_2O_2 + H$$

$$H_2O_2 + M \rightarrow OH + OH + M$$

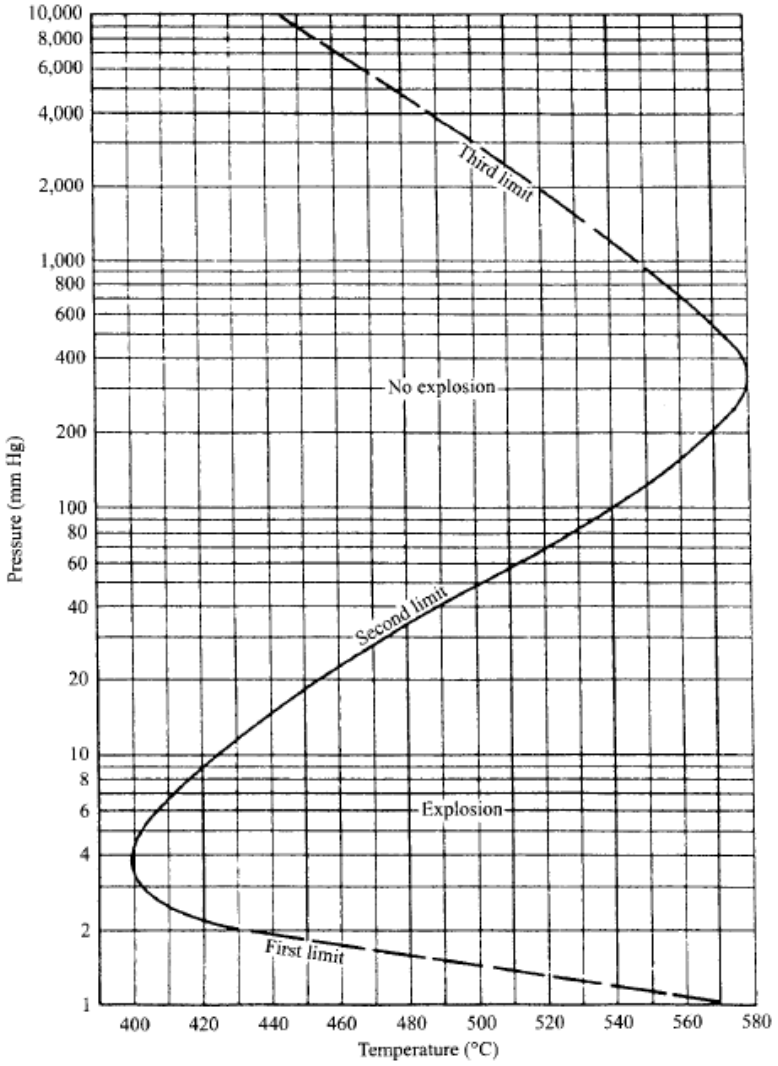


Figure 5.1 Explosion limits for a stoichiometric hydrogen–oxygen mixture in a spherical vessel.
SOURCE: From Ref. [2]. Reprinted by permission of Academic Press.

$$\frac{d[\mathrm{H}]}{dt} = -k_1[\mathrm{H}][\mathrm{O}_2] + k_2[\mathrm{O}][\mathrm{H}_2] + k_3[\mathrm{OH}][\mathrm{H}_2] - k_9[\mathrm{H}][\mathrm{O}_2][\mathrm{M}]$$

$$\frac{d[\mathrm{O}]}{dt} = k_1[\mathrm{H}][\mathrm{O}_2] - k_2[\mathrm{O}][\mathrm{H}_2]$$

$$\frac{d[\mathrm{OH}]}{dt} = k_1[\mathrm{H}][\mathrm{O}_2] + k_2[\mathrm{O}][\mathrm{H}_2] - k_3[\mathrm{OH}][\mathrm{H}_2].$$

Assuming O and OH in steady state

$$k_2[\mathrm{O}][\mathrm{H}_2] = k_1[\mathrm{H}][\mathrm{O}_2]$$

$$k_3[\mathrm{OH}][\mathrm{H}_2] = k_1[\mathrm{H}][\mathrm{O}_2] + k_2[\mathrm{O}][\mathrm{H}_2] = 2k_1[\mathrm{H}][\mathrm{O}_2].$$

$$\frac{d[\mathrm{H}]}{dt} = 2k_1[\mathrm{H}][\mathrm{O}_2] - k_9[\mathrm{H}][\mathrm{O}_2][\mathrm{M}]$$

$$= (2k_1 - k_9[\mathrm{M}])[\mathrm{H}][\mathrm{O}_2],$$



$$2k_1 = k_9[\mathrm{M}]$$

H2-Explosion Limit: Experiment vs Calculation

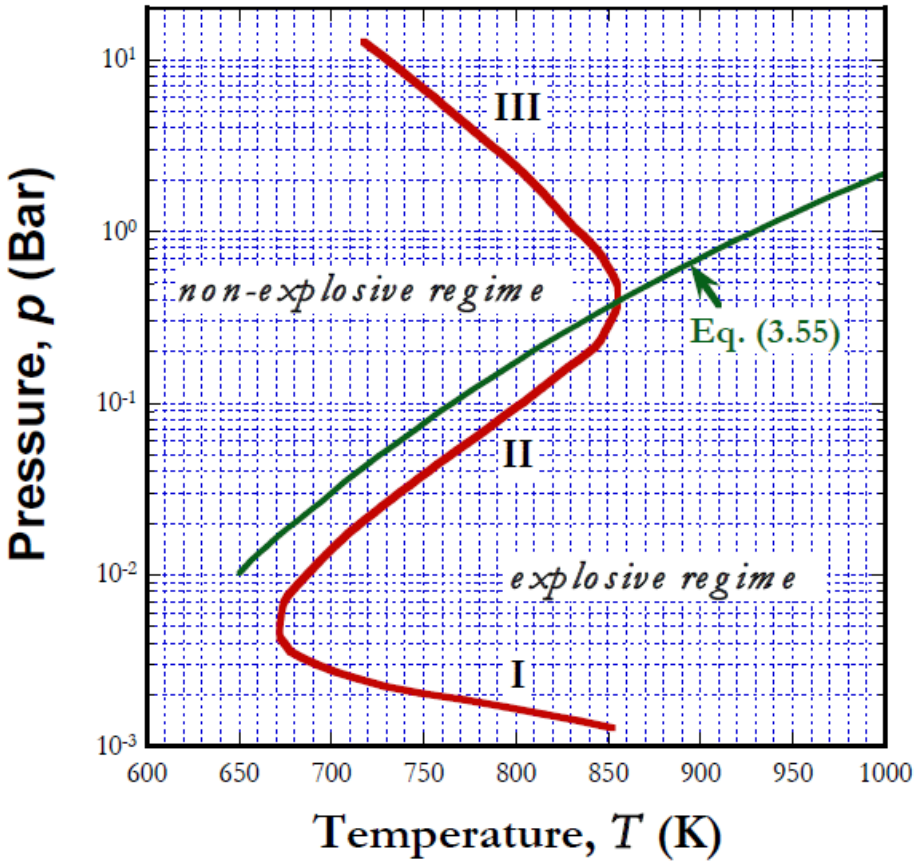


Figure 3.6 Explosion limits of hydrogen.

The complete H2-O2 Mechanism

No.	Reaction	B[cm, mol, s]	α	E_a(kcal/mol)
	H_2-O_2 Chain Reactions			
(1)	$H + O_2 \rightleftharpoons O + OH$	1.9×10^{14}	0	16.44
(2)	$O + H_2 \rightleftharpoons H + OH$	5.1×10^{04}	2.67	6.29
(3)	$OH + H_2 \rightleftharpoons H + H_2O$	2.1×10^{08}	1.51	3.43
(4)	$O + H_2O \rightleftharpoons OH + OH$	3.0×10^{06}	2.02	13.40
	H_2-O_2 Dissociation/Recombination			
(5)	$H_2 + M \rightleftharpoons H + H + M$	4.6×10^{19}	−1.40	104.38
(6)	$O + O + M \rightleftharpoons O_2 + M$	6.2×10^{15}	−0.50	0
(7)	$O + H + M \rightleftharpoons OH + M$	4.7×10^{18}	−1.0	0
(8)	$H + OH + M \rightleftharpoons H_2O + M$	2.2×10^{22}	−2.0	0
	Formation and Consumption of HO_2			
(9)	$H + O_2 + M \rightleftharpoons HO_2 + M$	6.2×10^{19}	−1.42	0
(10)	$HO_2 + H \rightleftharpoons H_2 + O_2$	6.6×10^{13}	0	2.13
(11)	$HO_2 + H \rightleftharpoons OH + OH$	1.7×10^{14}	0	0.87
(12)	$HO_2 + O \rightleftharpoons OH + O_2$	1.7×10^{13}	0	−0.40
(13)	$HO_2 + OH \rightleftharpoons H_2O + O_2$	1.9×10^{16}	−1.00	0
	Formation and Consumption of H_2O_2			
(14)	$HO_2 + HO_2 \rightleftharpoons H_2O_2 + O_2$	4.2×10^{14}	0	11.98
		1.3×10^{11}	0	−1.629
(15)	$H_2O_2 + M \rightleftharpoons OH + OH + M$	1.2×10^{17}	0	45.50
(16)	$H_2O_2 + H \rightleftharpoons H_2O + OH$	1.0×10^{13}	0	3.59
(17)	$H_2O_2 + H \rightleftharpoons H_2 + HO_2$	4.8×10^{13}	0	7.95
(18)	$H_2O_2 + O \rightleftharpoons OH + HO_2$	9.5×10^{06}	2.0	3.97
(19)	$H_2O_2 + OH \rightleftharpoons H_2O + HO_2$	1.0×10^{12}	0	0
		5.8×10^{14}	0	9.56

Explosion Limits of Hydrocarbons

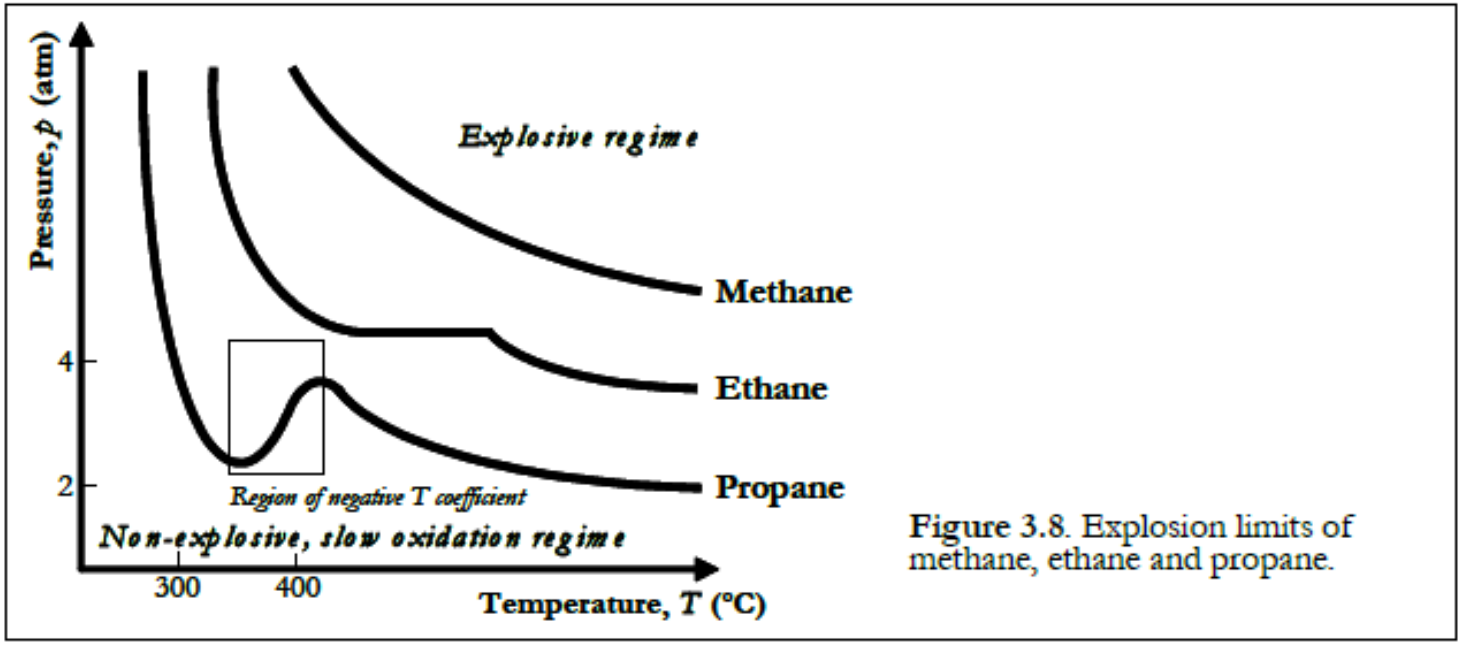


Figure 3.8. Explosion limits of methane, ethane and propane.

CO – O2 System

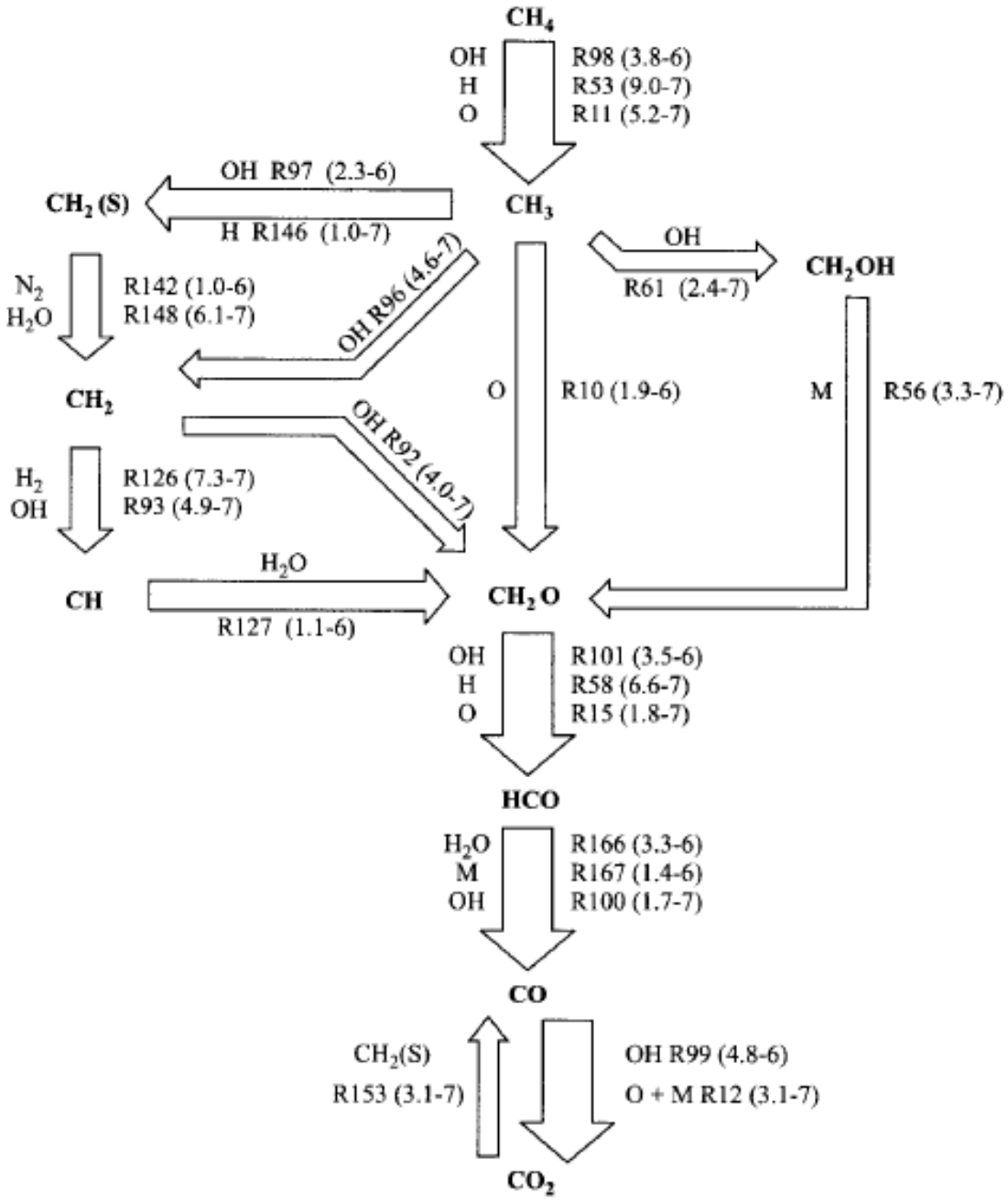
Oxidation of CO

(1)	$CO + O + M \rightleftharpoons CO_2 + M$	2.5×10^{13}	0	−4.54
(2)	$CO + O_2 \rightleftharpoons CO_2 + O$	2.5×10^{12}	0	47.69
(3)	$CO + OH \rightleftharpoons CO_2 + H$	1.5×10^{07}	1.3	−0.765
(4)	$CO + HO_2 \rightleftharpoons CO_2 + OH$	6.0×10^{13}	0	22.95

Reaction (2) is slow due to the large activation energy
Reaction (3) can be much faster:
OH, H, O radicals are equilibrium products of H2O

Dryer CO oxidation is slow
Wet CO (CO-O2-H2O) oxidation is much faster, due to the catalytic effect of H2O

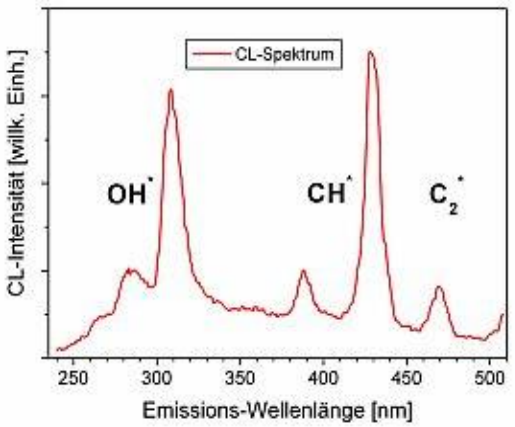
Methane – O2, High Temperature



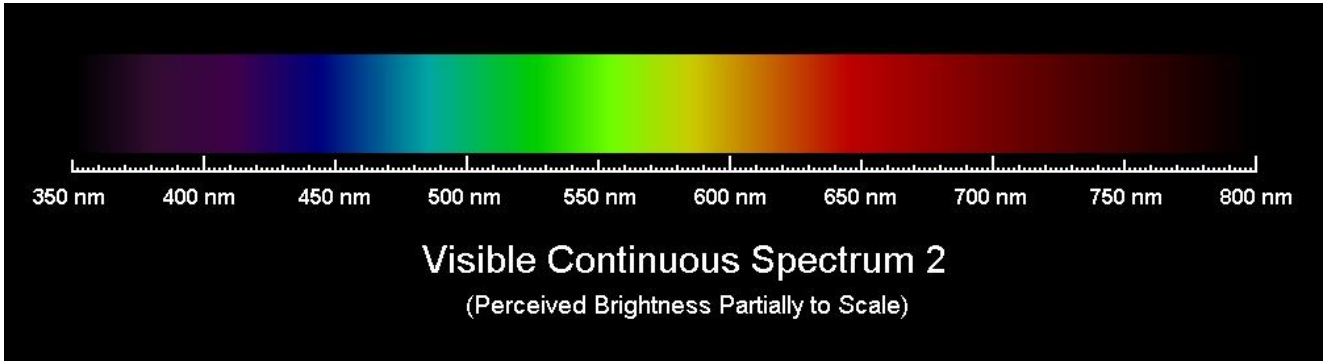
Chemiluminescence



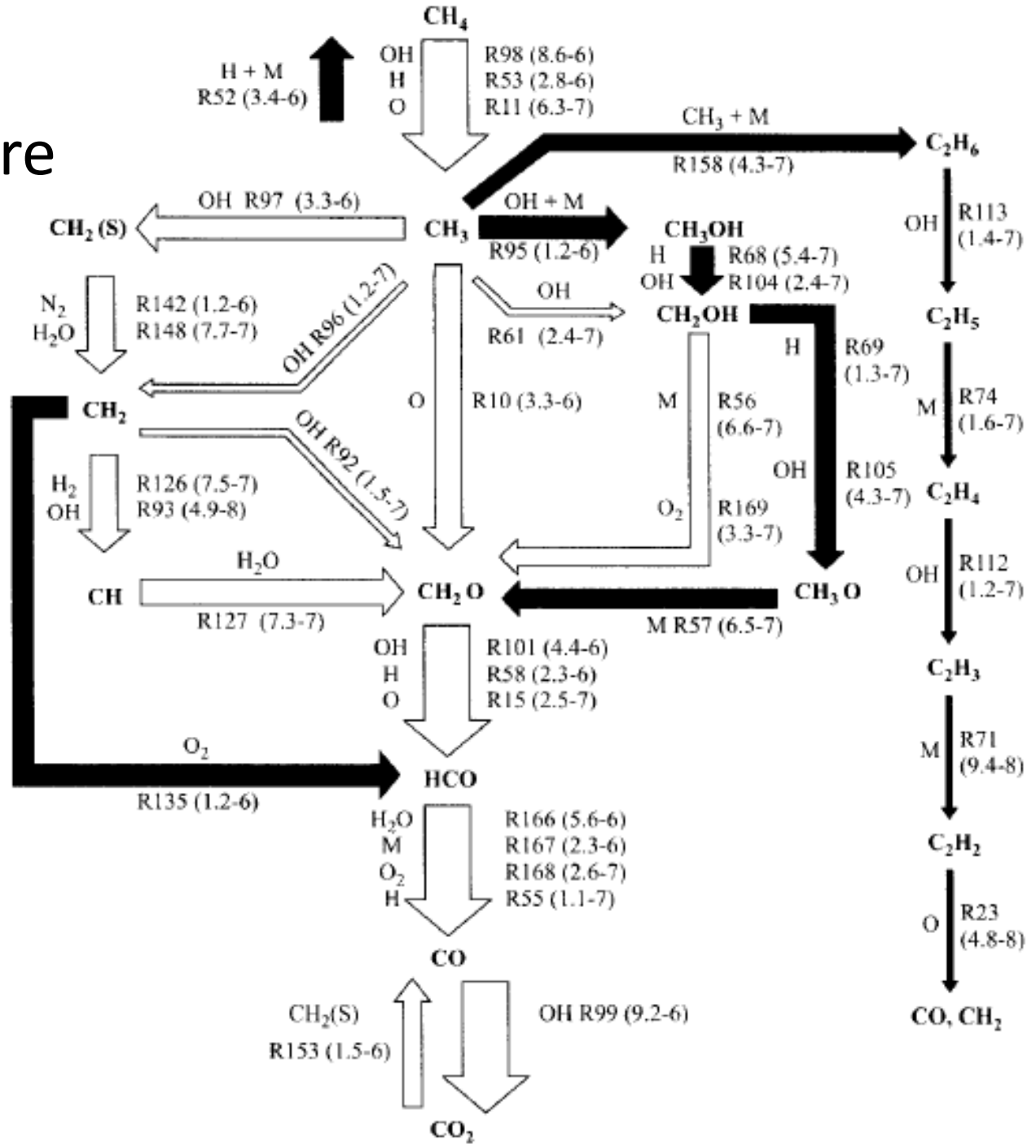
chemiluminescence from a Bunsen burner flame



- Responsible species: electronically excited OH* (310 nm), CH* (388, 431 nm) and C2* (473 nm)



Methane – O2, Low Temperature



GRI Mech 2.11

49 species, 279 reactions

Table 5.3 Complex methane combustion mechanism (GRI Mech 2.11) [10]

No.	Reaction	Forward Rate Coefficient[a]		
		A	*b*	*E*
	C–H–O Reactions			
1	$O + O + M \rightarrow O_2 + M$	1.20E + 17	−1.0	0.0
2	$O + H + M \rightarrow OH + M$	5.00E + 17	−1.0	0.0
3	$O + H_2 \rightarrow H + OH$	5.00E + 04	2.67	6,290
4	$O + HO_2 \rightarrow OH + O_2$	2.00E + 13	0.0	0.0
5	$O + H_2O_2 \rightarrow OH + HO_2$	9.63E + 06	2.0	4,000
6	$O + CH \rightarrow H + CO$	5.70E + 13	0.0	0.0
7	$O + CH_2 \rightarrow H + HCO$	8.00E + 13	0.0	0.0
8[b]	$O + CH_2(S) \rightarrow H_2 + CO$	1.50E + 13	0.0	0.0
9[b]	$O + CH_2(S) \rightarrow H + HCO$	1.50E + 13	0.0	0.0
10	$O + CH_3 \rightarrow H + CH_2O$	8.43E + 13	0.0	0.0
11	$O + CH_4 \rightarrow OH + CH_3$	1.02E + 09	1.5	8,600
12	$O + CO + M \rightarrow CO_2 + M$	6.02E + 14	0.0	3,000
13	$O + HCO \rightarrow OH + CO$	3.00E + 13	0.0	0.0
14	$O + HCO \rightarrow H + CO_2$	3.00E + 13	0.0	0.0
15	$O + CH_2O \rightarrow OH + HCO$	3.90E + 13	0.0	3,540
16	$O + CH_3OH \rightarrow OH + CH_3O$	1.00E + 13	0.0	0.0

… …

277	$NH_3 + H \rightarrow NH_2 + H_2$	5.40E + 05	2.4	9,915
278	$NH_3 + OH \rightarrow NH_2 + H_2O$	5.00E + 07	1.6	955
279	$NH_3 + O \rightarrow NH_2 + OH$	9.40E + 06	1.94	6,460

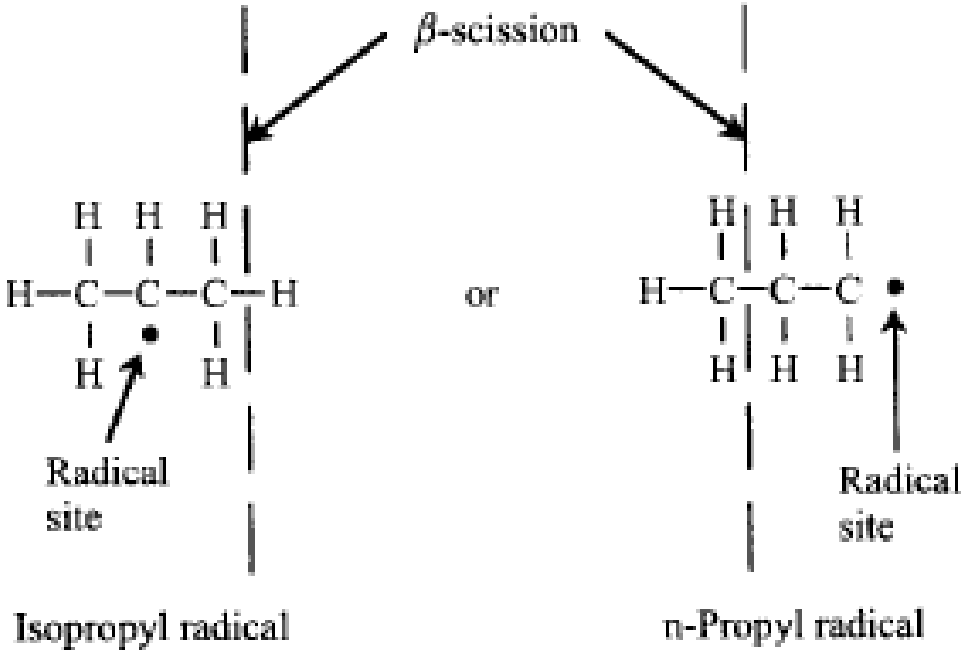
Oxidation of Higher Hydrocarbons

- Radical formation, e.g.

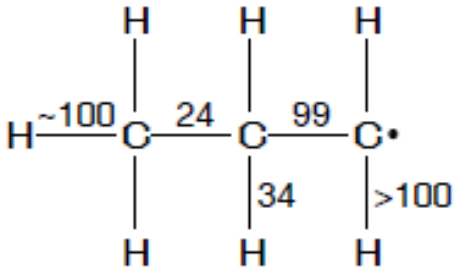
$$C_3H_8 + M \rightarrow C_2H_5 + CH_3 + M.$$

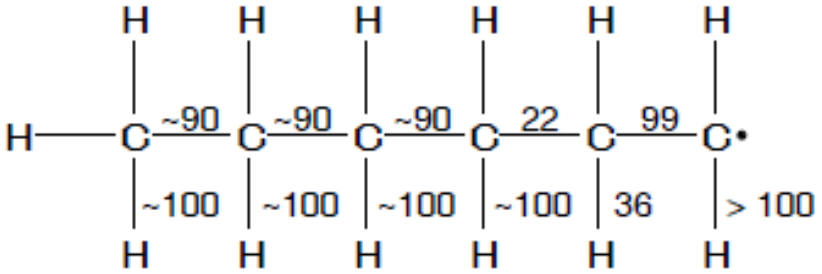
$$C_3H_8 + OH \rightarrow C_3H_7 + H_2O,$$

- β-scission

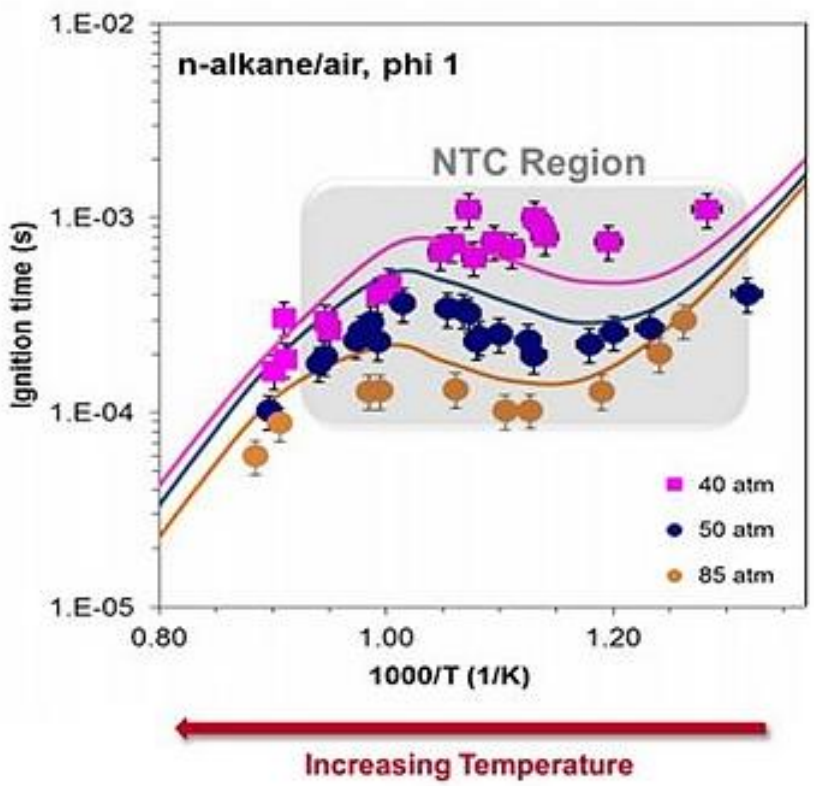


Examples of BDE (kcal/mole)





Low-Temperature Chemistry: Negative Temperature Coefficients (NTC)



$$RH + O_2 \rightarrow R + HO_2 \quad ,$$

$$R + O_2 + M \rightleftharpoons RO_2 + M$$

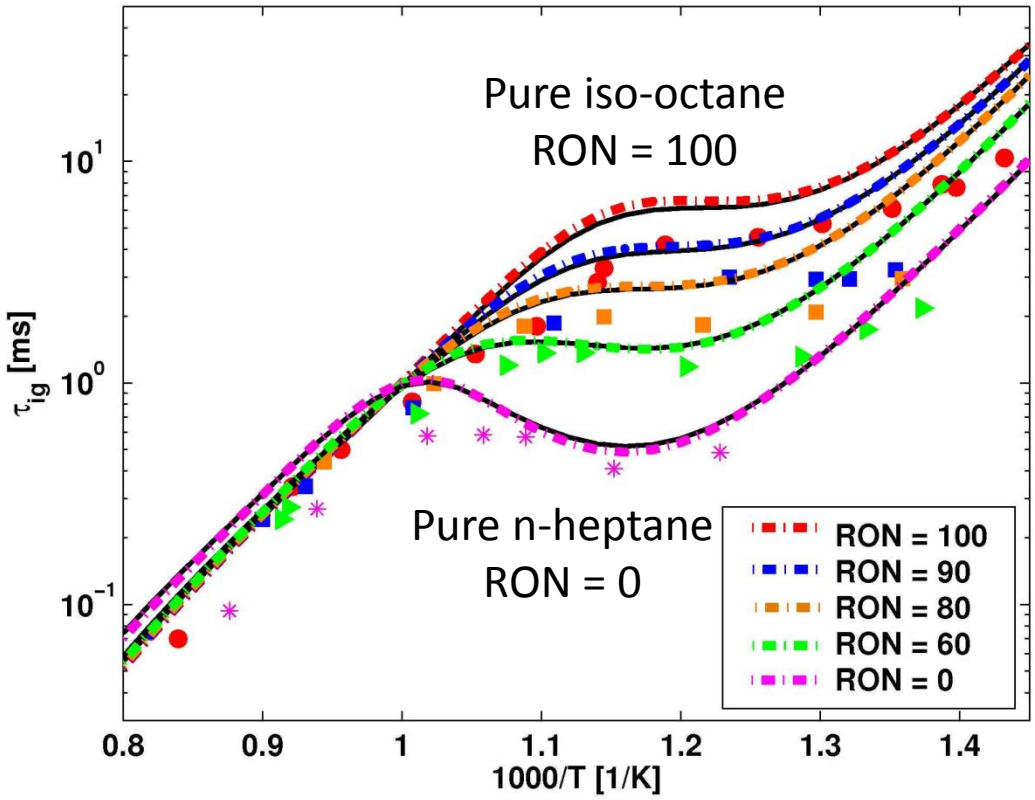
$$R + O_2 \rightarrow \text{olefin} + HO_2$$

$$RO_2 + RH \rightarrow RO_2H + R$$

$$RO_2H + M \rightarrow RO + OH + M$$

$$RH + (OH, HO_2, RO) \rightarrow R + (H_2O, H_2O_2, ROH)$$

Effect of Octane Number on Ignition Delay



RON: research octane number,

using mixtures of iso-octane and n-heptane, run engine at 600RMP

MON: measured at 900 RPM, more severe conditions

MON and RON for Typical Fuels

TABLE 1
OCTANE NUMBERS OF PURE HYDROCARBONS

Hydrocarbon	Actual RON	Actual MON	Blending RON	Blending MON
PARAFFINS				
n-butane	93		113	114
n-pentane	62	62	62	67
2-methylbyutane	92	90	99	104
2,2-dimethylpropane	85	80	100	90
n-hexane	25	26	19	22
2,2-dimethylbutane	92	93	89	97
n-heptane	0	0	0	0
2,2-dimethylpentane	93	96	89	93
2,2,3-trimethylbutane	>100	>100	113	113
2,2,3-trimethylpentane	100	100	105	112
2,2,4-trimethylpentane	100	100	100	100
OLEFINS				
1-Pentane	91	77	152	135
2-Methyl-2 butene	97	85	176	141
3-Methyl-2 pentene	97	81	130	118
4-Methyl-2 pentene	99	84	130	128
2,2,4-trimethyl-1 pentene	>100	86	164	153
2,2,4-trimethyl-2 pentene	>100	86	148	139
AROMATICS				
Benzene	>100	>100	99	91
Toluene	>100	>100	124	112
O-Xylene	>100	>100	120	103
M-Xylene	>100	>100	145	124
P-Xylene	>100	>100	146	127
Ethylbenzene	>100	98	124	107
1,3,5-trimethylbenzene	>100	>100	171	137
Propylbenzene	>100	98	127	129
Isopropylbenzene	>100	99	132	124
NAPHTHENES				
Cyclopentane	101	85	141	141
Methylcyclopentane	91	80	107	99
Cyclohexane	83	77	110	97
Methylcyclohexane	75	71	104	84
O-Dimethylcyclohexane	81	79	85	83
M-Dimethylcyclohexane	67	64	67	65
P-Dimethylcyclohexane	68	65	66	63

NOx Chemistry

- Thermal NOx (Zeldovich mechanism)

$$O + N_2 \Leftrightarrow NO + N$$
$$N + O_2 \Leftrightarrow NO + O,$$
$$N + OH \Leftrightarrow NO + H.$$

- Typically important at high-T (e.g. > 2000K)
- Species naming

  NO: nitric oxide

  NO2: nitrogen dioxide

  N2O: nitrous oxide

N_2O Intermediate Mechanism

- Important reactions

$$O + N_2 + M = N_2O + M$$
$$H + \ N_2O = NO + NH$$
$$O + N_2O = NO + NO$$

- Typically important for fuel lean ($\phi < 0.8$) and low-T conditions

Prompt NO (Fenimore mechanism)

- Important reactions

$$CH + N_2 \Leftrightarrow HCN + N$$

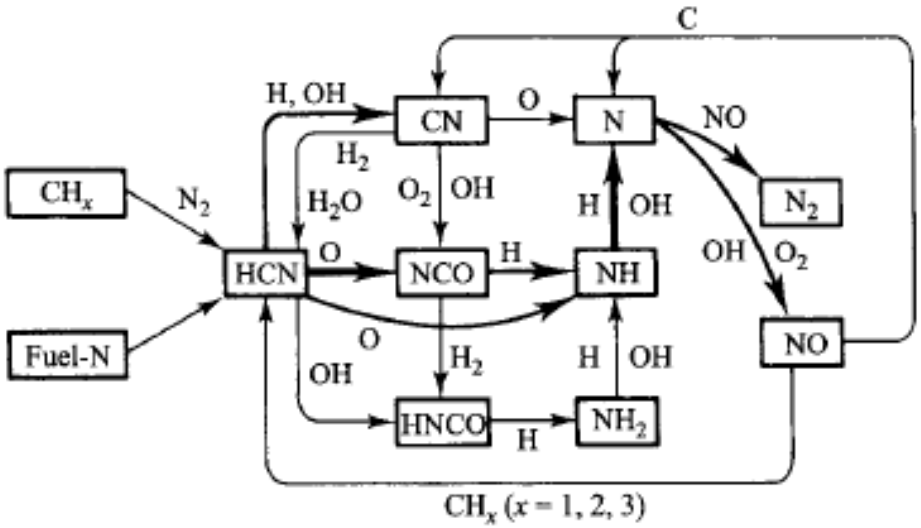
$$C + N_2 \Leftrightarrow CN + N,$$

$$HCN + O \Leftrightarrow NCO + H,$$

$$NCO + H \Leftrightarrow NH + CO,$$

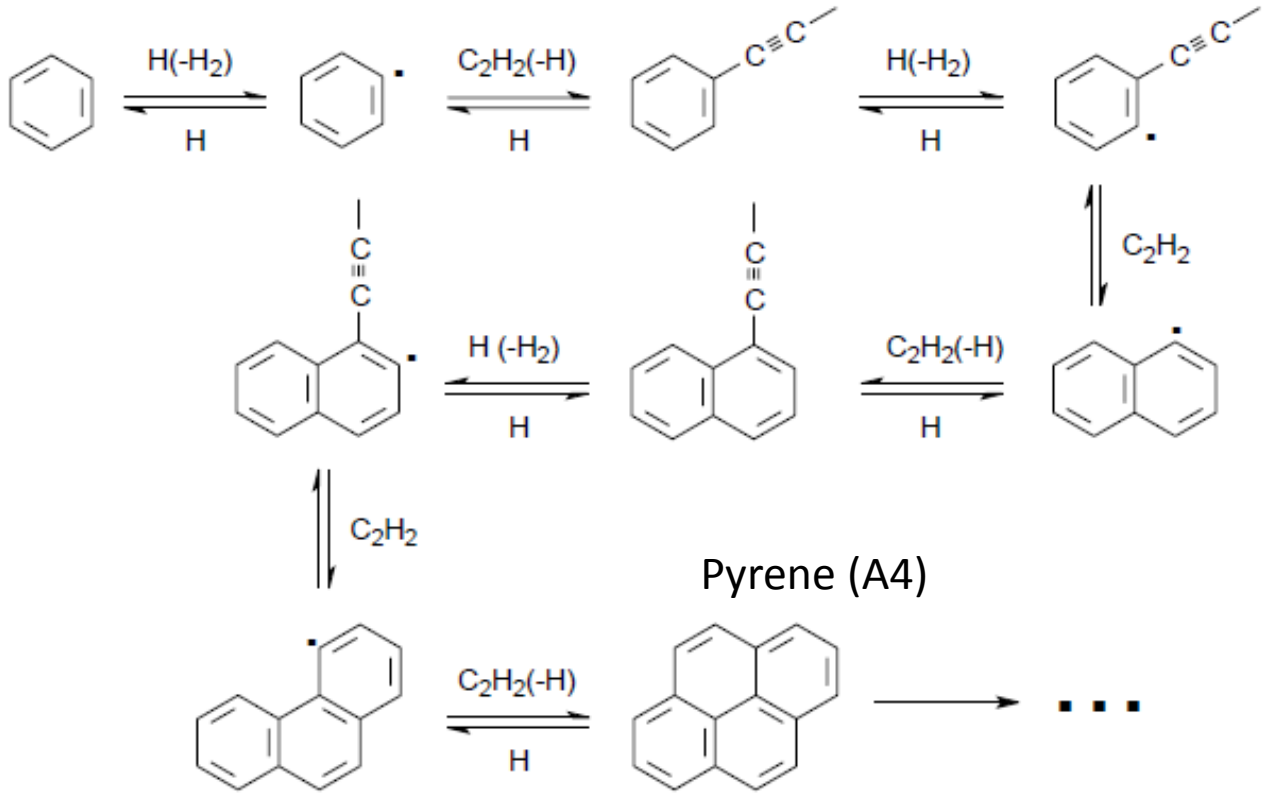
$$NH + H \Leftrightarrow N + H_2,$$

$$N + OH \Leftrightarrow NO + H.$$



- Typically important in flame fronts, where radicals (CH, CH2) are in peak concentrations

Soot Formation Through HACA



- HACA: hydrogen-abstraction C_2H_2-addition

Exceptions of Beta-Scission

